Andreas Happe

Security Researcher & Consultant

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Summary

Founder of offensive.one, PhD student at TU Wien researching the use of large language models for offensive security, and ISO 27001 auditor. Twenty years in security engineering, penetration testing and secure software development.

Experience

offensive.one GmbH, Founder — Vienna, Austria
2025 – present
1 year

Helping companies improve their security posture and secure software development practice ("left-shifting" security), with a focus on the intersection between AI and security.

- Zenity (since December 2025): contract research on AI agent security.

TÜV Süd, ISO 27001 Auditor — 2025 – present
1 year

TTTech Industrial Automation AG, Corporate Trainer — Vienna, Austria
2022 – 2023
1 year

Design and delivery of cybersecurity training for software developers.

Security Freelancer, Supporting Application Security — 2021 – 2025
4 years

Helping companies improve their security posture and secure software development practice ("left-shifting" security).

- dataspot. gmbh (2023–2024): improving the security posture and accompanying the ISO 27001 certification.

FH Technikum Wien, Lecturer — Vienna, Austria
2019 – 2023
4 years

Courses on Web Security, Secure Operating Systems and Web Application Security.

- Supervised ~10 master's theses, most completed with distinction.
- Team leader of the Linux and Web blue teams at the NATO Locked Shields Partner Event 2019 (2nd place).
- Author of the German textbook *Einführung in die Web Application Security* (2019, https://amzn.to/2ZqysuD).

CoreTEC GmbH, Senior Security Consultant — Vienna, Austria
2012 – 2022
10 years

Security assessments, penetration tests and secure software development training.

Austrian Institute of Technology, Software Engineer / Software Architect — Vienna, Austria
2006 – 2018
12 years

Twelve years on security research projects, first as a contractor (2006–2012), then as software architect (2013–2015) and software engineer (2015–2018).

- 2015–2018: Design, implementation and maintenance of privacy-preserving multi-cloud storage, identity management and data-sharing systems (PRISMACLOUD and CREDENTIAL, both Horizon 2020).
- 2012–2015: Architecture and implementation of the Archistar secure multi-cloud storage system.
- 2006–2012: Development and maintenance of the networking software of the SECOQC quantum key distribution network.

Ruby on Rails Freelancer, Developer — 2009 – 2015
Design, development and maintenance of Ruby on Rails-based web applications. — 6 years

TU Wien, Tutor — Vienna, Austria
Tutor for "Programming 101" and "Compiler Design" (Compilerbau). — 2004 – 2006, 2 years

Infotech GmbH, System Administrator — 2001 – 2007
Linux and Microsoft Windows systems. — 6 years

Volunteer

OWASP Foundation, Project and Chapter Leader — 2017 – present

- 2024–present: Founder and leader, OWASP Operational Technology (OT) Top 10.
- 2023–present: Project leader, OWASP Top 10 Proactive Controls.
- 2024–2026: Founder and leader, OWASP Chapter Klagenfurt.
- 2019–2023: Leader, OWASP Chapter Vienna.
- 2017: Top contributor, OWASP Mobile Security Testing Guide (MSTG).

Austrian Standards Institute, Standardization work — 2016 – 2026

- 2024–2026: ISO 42001 — AG001.42, mirror group ISO/CEN 42001.
- 2016–2019: ÖNORM A77.00 "Sichere Webapplikationen" — Austrian standard on development and maintenance of secure web applications.

Certificates

ISO 22301 Practitioner — 2026

GitHub Advanced Security Certified Practitioner — 2025

ISMS-Manager & -Auditor nach ISO 27001 — 2024

AWS Certified Developer — Associate — 2024

AWS Certified Solutions Architect — Associate — 2024

Certified OT Security Practitioner (COSP) — 2024

Certified Azure Red Team Professional (CARTP) — 2024

Certified Red Team Expert (CRTE) — 2024

Investigating Windows Endpoints (Gold) — 2023

Practical Network Penetration Tester (PNPT) — 2023

NIS-G Auditor für Kritische Infrastruktur — 2020

Offensive Security Certified Professional (OSCP) — 2015

Education

TU Wien, Computer Sciences — Usage of LLMs for Offensive Security — Vienna, Austria
Doctoral programme in Engineering Sciences, advised by Jürgen Cito. — 2022 – present

- GitHub AI Accelerator, cohort of 2024 (April–July): participating project hackingBuddyGPT, the research framework behind the PhD experiments.

TU Wien, Software Engineering & Internet Computing — Vienna, Austria, 2006 – 2009

- Master's thesis: *Agile Provenance* — automated transparent provenance gathering and analysis within Ruby on Rails. Supervisors: S. Dustdar, L. Juszczyk, H.-L. Truong.

TU Wien, Software & Information Engineering — Vienna, Austria, 2002 – 2006

HTBLVA Villach, EDV und Organisation — Villach, Austria, 1996 – 2001

Publications

Can LLMs Hack Enterprise Networks? Autonomous Assumed Breach Penetration-Testing Active Directory Networks — 2026
Andreas Happe, Jürgen Cito
doi.org/10.1145/3766895 (ACM Transactions on Software Engineering and Methodology (TOSEM); journal-first at ICSE 2026)

LLMs as Hackers: Autonomous Linux Privilege Escalation Attacks — 2026
Andreas Happe, Aaron Kaplan, Jürgen Cito
doi.org/10.1007/s10664-025-10758-3 (Empirical Software Engineering (EMSE); journal-first at ASE 2026)

Cochise: A Reference Harness for Autonomous Penetration Testing — 2026
Andreas Happe, Jürgen Cito
doi.org/10.1145/3832783.3834651 (ASE 2026, Tool Demonstrations Track, Munich, Germany)

Adversarial Bug Reports as a Security Risk in Language Model-Based Automated Program Repair — 2026
Piotr Przymus, Andreas Happe, Jürgen Cito
doi.org/10.1145/3793302.3793352 (MSR 2026)

ATLAS: Discovering Agent Strategies through LLM-Guided Abstraction and Automata Learning — 2026
Ignacio D. Lopez-Miguel, Andreas Happe, Jürgen Cito, Ezio Bartocci, Bettina Könighofer, Martin Tappler
doi.org/10.1145/3822455.3838777 (MODELS 2026, New Ideas and Emerging Results Track)

Towards Reliable Local Security Agents: Verifiable Post-Training for Linux Privilege Escalation — 2026
Philipp Normann, Andreas Happe, Jürgen Cito, Daniel Arp
arxiv.org/abs/2603.17673 (ACSAC 2026 (accepted))

The Ethics of Autonomous AI Agents for Offensive Security — 2026
Andreas Happe, Jürgen Cito, Jasmin Wachter
arxiv.org/abs/2607.20255 (FAIEMA 2026)

Enhancing Linux Privilege Escalation Attack Capabilities of Local LLM Agents — 2026
Benjamin Probst, Andreas Happe, Jürgen Cito
arxiv.org/abs/2604.27143 (RAISE Workshop at ESORICS 2026)

Recognition Without Mitigation: Ethical Frameworks in Autonomous Offensive-LLM Agent Research — 2026
Andreas Happe, Jürgen Cito
arxiv.org/abs/2506.08693 (AutonomousCyber Workshop at ESORICS 2026)

Benchmarking Practices in LLM-driven Offensive Security: Testbeds, Metrics, and Experiment Design 2025
Andreas Happe, Jürgen Cito
arxiv.org/abs/2504.10112 (arXiv preprint; presented at DeMeSSAI 2025, Venice, Italy)

On the Surprising Efficacy of LLMs for Penetration-Testing 2025
Andreas Happe, Jürgen Cito
arxiv.org/abs/2507.00829 (arXiv preprint)

Getting pwn'd by AI: Penetration Testing with Large Language Models 2023
Andreas Happe, Jürgen Cito
doi.org/10.1145/3611643.3613083 (ESEC/FSE 2023, Ideas, Visions and Reflections Track, San Francisco, USA)

Understanding Hackers' Work: An Empirical Study of Offensive Security Practitioners 2023
Andreas Happe, Jürgen Cito
doi.org/10.1145/3611643.3613900 (ESEC/FSE 2023, Industry Track, San Francisco, USA)

Unikernels for Cloud Architectures: How Single Responsibility can Reduce Complexity, Thus Improving Enterprise Cloud Security 2017
Andreas Happe, Bob Duncan, Alfred Bratterud
(COMPLEXIS 2017, Porto, Portugal)

The Archistar Secret-Sharing Backup Proxy 2017
Andreas Happe, Florian Wohner, Thomas Lorünser
(SECPID / ARES 2017, Calabria, Italy)

Enterprise IoT Security and Scalability: How Unikernels can Improve the Status Quo 2016
Bob Duncan, Andreas Happe, Alfred Bratterud
(IEEE/ACM UCC 2016, Shanghai, China)

Enhancing Cloud Security and Privacy: Time for a New Approach? 2016
Bob Duncan, Alfred Bratterud, Andreas Happe
(INTECH 2016, Dublin, Ireland)

Exchanging Database Writes with Modern Crypto 2016
Andreas Happe, Thomas Lorünser
(IARIA CYBER 2016, Venice, Italy)

Malicious Clients in Distributed Secret Sharing Based Storage Networks 2016
Andreas Happe, Stephan Krenn, Thomas Lorünser
(Security Protocols Workshop 2016, Brno, Czech Republic)

ARCHISTAR: Towards Secure and Robust Cloud Based Data Sharing 2015
Thomas Lorünser, Andreas Happe, Daniel Slamanig
(IEEE CloudCom 2015, Vancouver, Canada)

New Release of an Open Source QKD Software: Design and Implementation of New Algorithms, Modularization and Integration with IPSec 2013
Oliver Maurhart, Christoph Pacher, Andreas Happe, Thomas Lorünser, Cristina Tamas, Andreas Poppe, Momtchil Peev
(QCrypt 2013)

QKD Software Architecture and System Integration with Classical Communication Infrastructure 2012
Oliver Maurhart, Christoph Pacher, Andreas Happe, Thomas Lorünser, Cristina Tamas, Andreas Poppe, Momtchil Peev
(QCrypt 2012)

The SECOQC Quantum Key Distribution Network in Vienna 2009
Momtchil Peev, Christoph Pacher, Romain Alléaume, et al.
doi.org/10.1088/1367-2630/11/7/075001 (New Journal of Physics)

Projects

hackingBuddyGPT 2023 – present
Original author. A research framework for building LLM-driven offensive-security agents in 50 lines of Python or less; backs several of my papers and has grown into a community project.
- Selected for the GitHub AI Accelerator 2024.

Cochise 2025 – present
Open-source reference harness for autonomous assumed-breach penetration testing of Active Directory networks, evaluated against the GOAD testbed. Accompanying paper in ACM TOSEM, tool paper at ASE 2026.

OWASP Operational Technology (OT) Top 10 2024 – present
Project founder and leader. Awareness document listing the most critical security risks in operational technology.

OWASP Top 10 Proactive Controls 2023 – present
Project leader. The developer-facing list of the ten security techniques every software project should build in.

PRISMACLOUD (Horizon 2020) 2015 – 2018
Design, development and maintenance of the privacy-preserving multi-cloud storage prototype. EU Horizon 2020 project coordinated by AIT (grant 644962).

CREDENTIAL (Horizon 2020) 2015 – 2018
Development of the data-sharing and identity-management components of a secure cloud identity wallet based on proxy re-encryption. EU Horizon 2020 project coordinated by AIT (grant 653454).

Archistar 2012 – 2015
Architecture and implementation of a secure multi-cloud storage system combining Byzantine fault tolerance with secret sharing. AIT research project funded by the Austrian BMVIT.

SECOQC 2006 – 2012
Networking components (Linux, Python, C) of the first metropolitan quantum key distribution network, demonstrated in Vienna in 2008; afterwards maintenance and further development of the open-source QKD software stack.

Skills

Security Engineering

Secure Software Engineering

Software Development

Languages

German
Native language

English
Full professional proficiency